Effective Upcycling of PET Plastics

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Rationale

Plastics, right now, are manufactured at a much greater rate than they can be reused. In fact, less than 10% of plastics are recycled worldwide (Korley et. al 2021). Because of the large amount of waste the industry produces, there is a great need for a circular economy in that regard.

Recycling plastic may help reduce the waste produced, but as Pilapitiya et. al (2024) notes, "For plastics and other classes of polymers to positively contribute as renewable resources to the economy, a shift toward circularity is needed." Rather than recycling plastic to make new, lower quality plastic, a shift from plastic polymer chains to a higher value material is needed for a circular economy. **To fix this issue, we aim to find a sustainable and affordable reaction via catalysis to depolymerize polyethylene terephthalate (PET),** which comprises most plastics. The process should create Ethylene glycol, a resource that would otherwise be distilled from crude oil.

Recent developments in PET upcycling have broken down PET into carbon capturing solids (Poderyte et. al 2025). While methods alternative to catalysis provide useful ways to use excess PET, they don't contribute to a circular economy that would stop the issue of plastic production altogether. When considering an industrial application of production however, limiting factors arise in methods involving catalysis. Minimal products and expensive equipment are not sustainable long-term (Gao 2010). To scale our reaction, we will maintain effective and sustainable conditions.

With one of the largest industries in the world producing some of the most resistant and non-decomposable or sustainable materials, people using everyday plastics should not have to worry about if their plastic is actually going to be recycled. Currently, the management of plastic waste predominantly relies on landfill disposal (Kwon et. al 2024). Plastic can only be recycled so many times before it loses more value than can be used in recycled products. **With upcycled materials, throwing a bottle away would mean another one would sustainably take its place.**

Criteria

Design Criteria	Goal	Actual
Use a working catalyst	Meet requirement	MET GOAL ✔
Be able to react PET with a hydrogen donor	Meet requirement	MET GOAL ✔
Be able to analyze reaction and products	Meet requirement	MET GOAL ✔
Produce upcycled products	>30% products	PARTIALLY MET GOAL ✔
Cost	<$300	<$150 ✔

Materials & Equipment

- 5 plastic PET plastic cup lids
- 25g container of zinc acetate
- 1L methanol container
- Spectrophotometer
- Hot plate
- Scale

Methanol

*Photo taken by Sigma Aldrich

Zinc acetate

*Photo taken by Sigma Aldrich

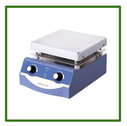
Spectrophotometer

*Photo taken by Vernier

Plastic cup lids

*Photo taken by ULINE

***Picture shown does not correspond to brand used in reaction**



Hot plate

*Photo taken by ATO

Scale

*Photo taken by U.S. Solid

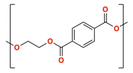
Cost Effective Reaction

The highest cost in our reaction was the **zinc acetate catalyst**. On a large scale, however, the catalyst could be **reused for each reaction** and the cost would instead be limited to methanol. Even though methanol is the limiting reactant and has twice the molar equivalent as the PET, methanol is a very affordable material that would cost effective on an industrial scale.

Research Goal:
Find alternative upcycled products of polyethylene terephthalate through catalysis

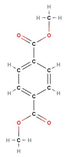
$$C_{10}H_8O_4\ (s) + 2\ CH_3OH\ (l) \Longrightarrow C_{10}H_{10}O_4\ (s) + C_2H_6O_2\ (l)$$

Polyethylene terephthalate

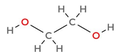


Methanol

Dimethyl terephthalate



Ethylene glycol



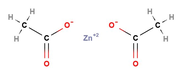
*Chemical figures created by MolView

This is the chemical formula and model for our reaction. **$C_{10}H_8O_4$ or polyethylene terephthalate** (PET) is the plastic. **CH_3OH or methanol** is the solvent and reactant with the PET. **$C_{10}H_{10}O_4$ or dimethyl terephthalate (DMT)** is a polymer similar to PET and is a component used in the production of many plastics, including PET. **$C_2H_6O_2$ or ethylene glycol (EG)** is a liquid used heavily in the production of PET plastics on an industrial scale.

The catalyst used is **zinc acetate**, it was used because of its **ability to react with organic molecules**, which both reactants are. The goal of the catalyst is to **lower energy** needed to react the PET with methanol. The catalyst is preserved after the reaction and can be reused once filtered out of the solution.

The product from the reaction results in a solution of white solid suspended in a clear liquid. The products are compared to the intended **DMT** and **EG** products to determine the success of the reaction. The experiment followed the chemical formula in measuring products and molar equivalents in measuring reactants.

Zinc acetate



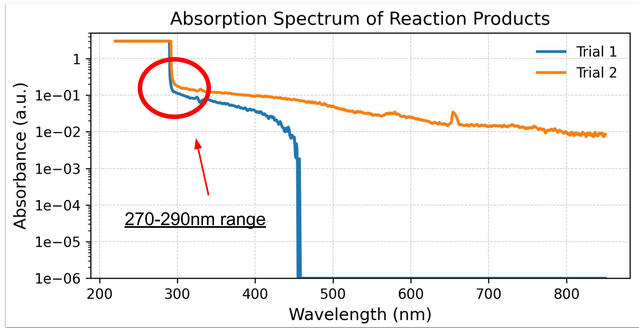
The figure to the left is the zinc acetate catalyst used in the reaction. The **Zn^{2+}** ion allows for the reaction to operate at much lower temperatures by **speeding up the reaction.**

Data & Analysis

The graph shows absorbance of the produced solution across the UV and visual spectrum of light derived by using a spectrophotometer. The graph is on a **logarithmic scale** to emphasize changes in absorption.

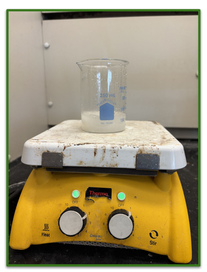
By interpreting this graph we can see that there is a big change in absorption consistent with what we would expect with DMT because of the indole ring in the chemical structure that absorbs light from the **270-290 nm range.** We can see that same drop in the same range on the graph. This consistency points towards the likelihood that **DMT was really produced from the reaction.**

EG does not absorb rays from the UV and visual spectrum. This means that the presence of **EG** as the solute can not be confirmed through spectroscopy. To point towards the likelihood of the product, we would instead defer to the high boiling point and other qualitative data consistent with **EG.**



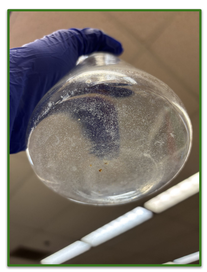
*Values below 0 are set to 1e-06 to account for errors in spectroscopy
**Figure created by researcher

This is the roughly shredded **PET** from trial 1 placed inside a 100 ml beaker with the **zinc acetate catalyst** added, as seen by the white powder at the bottom. The simple preparation of the trial highlights the viability it poses when reacting with massive amounts of **PET**.



This is the reaction process. A **diluted methanol solution** is added throughout the reaction and the stirring stick is turned with the hot plate. The reaction ran for 7 hours under these conditions. Due to methanol and water evaporation, the total amount of methanol used could not be determined.

This is the filtration process of the solution after the reaction was completed. Due to the limited time spent reacting, much of the plastic was left over. This filtration process removed the **PET** and the **zinc acetate**, which allowed for the desired products to settle and form into solids. We washed using cold methanol.



This is the solution produced after the filtration. **Visible white solids** can be seen suspended in the solution after chilling for 3 hours in an ice bath. This solution was prepared for the **spectroscopy**.

This is the **spectroscopy** performed with the two solutions produced after both trials were completed. Using software connected to the device, we were able to collect the absorption data across the visible and UV light spectrum.

***Picture above were taken by researcher**

Procedure

Step 1. Set up:

The PET plastic was shredded and measured alongside the zinc acetate catalyst into a 100ml beaker. After placing the beaker onto a prepared hot plate, a stirring rod was placed into the beaker. Methanol was diluted with water before the reaction.

Step 2. Reaction:

The reactants were placed on a hot plate set to 55°C for 7 hours with diluted methanol added erratically throughout both of the trials performed. The remaining plastic and zinc acetate were filtered out after the reaction.

Step 3. Data Collection:

The produced solutions were measured and used in a spectroscopy to determine absorption of light across the UV and visual light spectrum

Step 4. Data Analysis:

The data observed from the spectroscopy was graphed using a custom matplotlib python script. Consistencies in absorption were noted in correlation with the desired products.

Step 5. Scalability:

Given the data collected from the reaction, the final step is to compare the conducted upcycling process with current recycling processes. Alternative solutions were compared with the energy cost of our reaction to determine viability.

Conclusions & The Future

Our reaction indicates that **PET** plastic can be upcycled using a low cost and low energy method.

By analyzing products created, we can infer that **DMT** and **EG** were created successfully. Given the conditions at 55°C for the reaction, this points towards a **low cost alternative to traditional recycling** that can bolster a circular economy in plastic production. With more time and resources this study could both confirm the exact products created and complete more trials under common industrial conditions to determine viability. Energy use could also be measured and compared with industry standards in pyrolysis and other methods of plastic upcycling.

This process of upcycling yields promising results that can be scaled into **viable options** to actively **work against plastic pollution** and close the linear economy of plastic resulting in millions of tons of plastic being tossed into the ocean.

Bibliography

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